

Natural Language processing

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Home work - 1

Q.2. Manual BPE on a toy corpus.

2.1 using the same corpus from class:

low low low low low lowest lowest newer newer
newer newer newer wider wider wider new new

1. Add the end-of word marker _ and write the initial Vocabulary (characters + _).

Ans: Corpus with _

Each word is split into characters + _ :

L O w _

L O w _

L O w _

L O w _

L O w _

L O w e s t _

L O w e s t _

n e w e r _

n e w e r _

n e w e r _

n e w e r _

n e w e r _

n e w e r _

w i d e r _

w i d e r _

w i d e r _

n e w _

n e w _

Initial Vocabulary

{ l, o, w, e, s, t, n, r, i, d, _ }

2. Compute bigram counts and perform the first three merges by hand:

• Step ①: First Merge

(l o) $\rightarrow$ lo

updated corpus snippet (2 lines)

lo w _

lo w e s t _

New token

lo

updated Vocabulary

{l, o, w, e, s, t, n, r, i, d, _, lo}

• Step ②: Second Merge

after step 1, the most frequent bigram becomes: (lo w)

Merge $\rightarrow$ (lo w) $\rightarrow$ low

updated corpus snippet

low _

low e s t _

New token

low

updated Vocabulary

{l, o, w, e, s, t, n, r, i, d, _, lo, low}

• Step ③: Third merge

Now the most frequent pair is:

(low _)

Merge

(low _) $\rightarrow$ low _

updated corpus snippet:

low_

lowest_

New tokens:

low_

updated Vocabulary:

{l, o, w, e, s, t, n, r, i, d, -, lo, low, low_}.